Task 05 – Operators and Operator Precedence

1. Objective

To understand C++ operators, operator precedence, associativity, and their application in calculating metro fares, peak-hour surcharges, and passenger concessions.

2. Types of Operators

Operator Type	Symbols	Application
Arithmetic	+, –, *, /, %	Fare calculation
Relational	==, !=, >, <, >=, <=	Balance comparison
Logical	&&, ||, !	Journey validation
Assignment	=, +=, -=	Balance updates
Increment/Decrement	++, --	Journey counter
Conditional	?:	Fare selection

3. Operator Precedence

Priority	Operators	Associativity
1	++, -- (postfix)	Left to Right
2	++, -- (prefix), !	Right to Left
3	*, /, %	Left to Right
4	+, –	Left to Right
5	<, >, <=, >=	Left to Right
6	==, !=	Left to Right
7	&&	Left to Right
8	||	Left to Right
9	?:	Right to Left
10	=, +=, -=	Right to Left

4. Increment Operators

- i++ (post-increment): Uses the current value before increasing it.
- ++i (pre-increment): Increases the value before using it.
- i = i++ should be avoided because it is an unsafe and confusing modification pattern; use i++ or ++i separately.

5. Division and Modulus

C++ integer division truncates towards zero.

Expression	**Result**
10 / 3	3
10 % 3	1
-10 / 3	-3
-10 % 3	-1

6. Peak-Hour Condition

(hour >= 8 && hour < 10) ||
(hour >= 17 && hour < 20)

This checks both peak-hour periods: 8 AM–10 AM and 5 PM–8 PM. Parentheses make the intended grouping clear and prevent logical mistakes.

7. Alternative Approaches and Complexity

Approach	**Advantage**	**Limitation**
Direct logical expression	Simple and readable	Conditions must be maintained

Lookup table	Fast peak-hour lookup	Requires extra storage
Conditional operator	Compact fare selection	Can reduce readability if nested

Both direct checking and lookup-table checking take O(1) time. A 24-hour Boolean lookup table requires O(24) space, which is constant space.

8. Result

C++ operators and precedence rules help implement accurate fare calculations, peak-hour detection, balance updates, and passenger concessions. Proper parentheses and clear expressions improve program correctness and readability.